Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-Informed Neural-Network Optimization Framework — Final Report

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PHYS 400 — Final Presentation

Why Two Elastic Constants Govern Structural Design

Every load-bearing part is sized by two numbers:

- $E = \sigma/\varepsilon$ — Young's modulus: stiffness (GPa).
- $\nu = -\varepsilon_{\text{lat}}/\varepsilon_{\text{axial}}$ — Poisson's ratio: transverse contraction.

Symmetry reduces the cubic stress–strain law $\boldsymbol{\sigma} = \mathsf{C}\,\boldsymbol{\varepsilon}$ to just three independent constants:

$$\mathsf{C} = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{44} \end{bmatrix}$$

E and ν follow from C_{11}, C_{12} — **the quantities this work targets from first principles**.

They set: beam deflection $\propto 1/E$, buckling $\propto E$, $\omega_n \propto \sqrt{E/\rho}$, $K = E/[3(1-2\nu)]$, toughness.

Stability (Born–Cauchy): $C_{11} > C_{12}$.

Representative structural applications

Application	Material	E (GPa)	ν
Aerospace fuselage	**Al**	**65–80**	**0.3**
Turbine disk	Inconel 718	200	0.29
Orthopaedic implant	Ti 6Al–4V	110	0.34
Automotive structure	Mg AZ91	45	0.35
Structural steel	Mild steel	200	0.30
MEMS resonator	Si (single xtal)	130	0.28

Same two constants span polymers ($\nu \to 0.4$) to steel — and fix whether a design survives load.

Interatomic potential. Closed-form approximation of the many-body potential energy $U(\{\mathrm{r}_i\})$; forces obtained as $\mathrm{F}_i = -\nabla_i U$.

Historical development:

- Pairwise (Lennard–Jones, Morse) — too crude for metals.
- Embedded Atom Method (1984) — good for FCC metals.
- **Modified EAM** (MEAM) [1]: adds angular terms — covers BCC, HCP and mixed bonding.

Density-functional theory (DFT). First-principles electronic-structure method (Kohn–Sham equations [2]); here a plane-wave + pseudopotential basis in Quantum Espresso [3].

Quantities computed:

- Cohesive energy E_{coh}
- Equilibrium lattice a_0, bulk modulus B
- Binary formation energies $E_{\mathrm{form}}(i, j)$
- Single-crystal elastic constants C_{11}, C_{12}

DFT: high accuracy, limited scale ($\leq 10^3$ atoms).
MEAM/MD: empirical, large scale ($\geq 10^6$ atoms).
DFT provides the reference dataset against which the MEAM potential is calibrated.

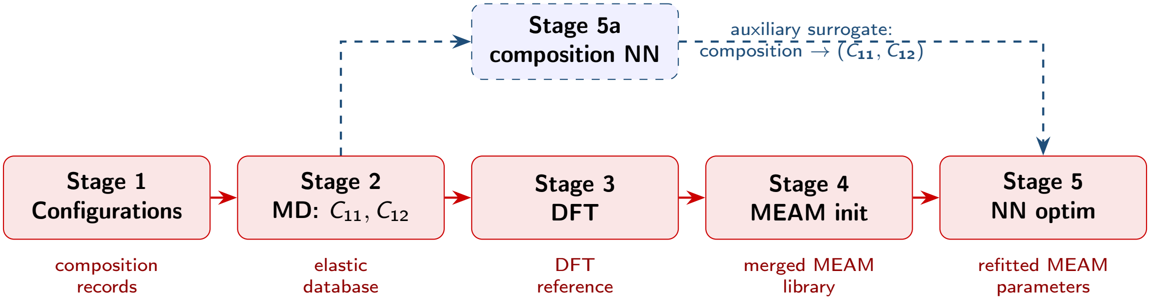
The MEAM Coverage Problem

Published MEAM libraries cover disjoint subsets of elements; no single file spans the nine elements specified by Al 7075.

Library	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	
Merged (this work)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Limitation of direct merging: cross-interaction parameters between elements drawn from distinct libraries are not defined; manual refitting does not scale to $\binom{12}{2} = 66$ binary pairs.

Pure machine-learning potentials [4–6]: require $10^4 - 10^5$ DFT configurations and forfeit the physical interpretability of MEAM.

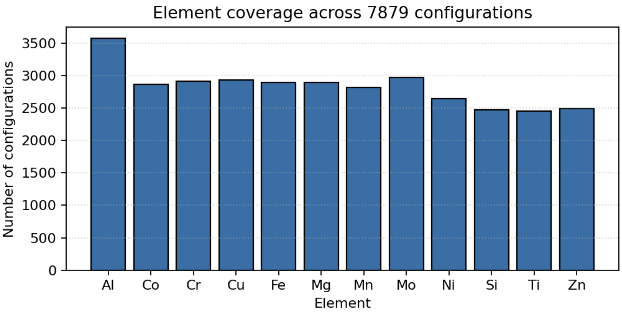


The four published libraries are merged into a single 12-element file; cross-terms are refitted by inverting an NN surrogate of the MEAM $\to (C_{11}, C_{12})$ map under L-BFGS, with single-element parameters anchored to DFT references.

MD = Molecular Dynamics; DFT = Density-Functional Theory; NN = Neural Network.

Sampling.
Uniform draws on the 12-element composition simplex (Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn);
Al-weighted to oversample the aluminium-alloy region.

Corpus statistics.
$\sim$7900 configurations generated;
7815 retained after the physicality filter ($E > 0$, $0 < \nu < 0.48$, $C_{11} > C_{12}$).



Element-occurrence distribution over the retained corpus.

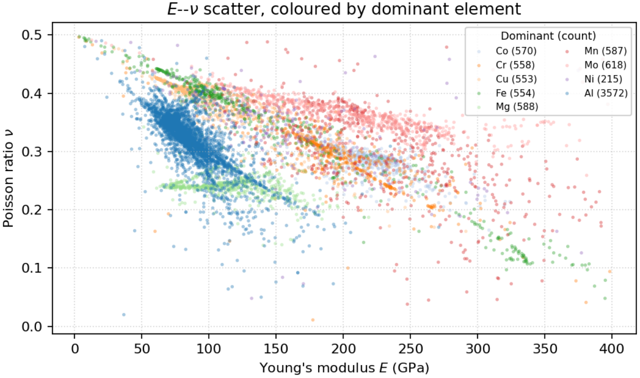
Implementation. [7]
Custom LAMMPS build (MEAM species limit raised $5 \to 20$).

Per-configuration protocol:

- ~ 4 nm cubic supercell
- Atom-type reassignment to composition $\{c_i\}$
- Anisotropic energy minimisation
- Axial strain $\delta = 10^{-3}$ applied along $\pm x, \pm y, \pm z$
- Central-difference $\to C_{11}, C_{12}$

	Median	Std.	Range
E (GPa)	103.6	69.1	3 – 398
ν	0.326	0.063	0.01 – 0.50

$N = 7815$ retained configurations.

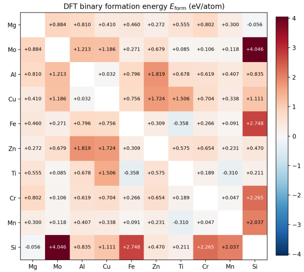


E–ν distribution, coloured by dominant element. Mg- and Zn-rich compositions cluster near $\nu \to 0.5$, motivating training on C_{11}, C_{12} rather than (E, ν).

Stage 3–4: DFT References → MEAM Initialisation

Element	Lat.	a_0 (Å)	E_{coh}	B (GPa)
Al	FCC	4.047	3.611	77.5
Cu	FCC	3.639	3.857	138.5
Si	DIAMOND	5.471	5.130	89.3
Ti	HCP	2.915	6.284	114.5
Zn	HCP	2.766	0.964	68.4
Cr	BCC	2.848	3.964	213.1
Fe	BCC	2.806	6.834	420.5
Mg	HCP	3.177	1.483	47.8
Mn	BCC	2.803	3.728	225.7
Mo	BCC	3.169	10.353	280.8

Elemental references (E_{coh} in eV/atom; Birch–Murnaghan fits).



Binary formation energies $E_{\mathrm{form}}(i,j)$ (eV/atom): negative = exothermic mixing, positive = limited solubility — these seed the MEAM cross-terms.

Stage 4 — single-element initialisation

Screening exponent from each DFT triple, $\alpha = a_0\sqrt{9B\Omega/E_{\mathrm{coh}}}$ (Ω: cell volume) → complete 12-element MEAM library (∼70 ms, cached). α-Mn EOS did not converge → relaxed-BCC approximation.

Stage 5a: Composition $\rightarrow (E, \nu)$ Neural Surrogate

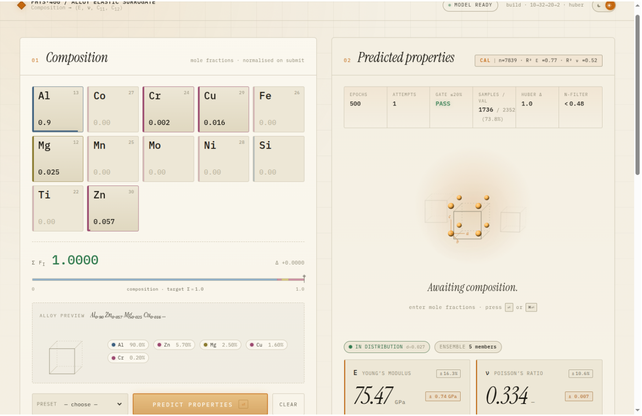
Architecture. $12 \to 32 \to 20 \to 2$ fully-connected MLP with ReLU activation; Huber loss on $\hat{C}_{11}$, $\hat{C}_{12}$; deep ensemble of five seeds [8] for epistemic uncertainty.

Result

Held-out (30 % split): $R_E^2 = 0.77$, $R_\nu^2 = 0.52$.

Al 7075 preset: $\hat{E} = 75.5\,\text{GPa}$, $\hat{\nu} = 0.334$ (published: 71–72 GPa, $\nu \approx 0.33$ — within $\sim 5\,\%$).

Distributed as a portable application image.



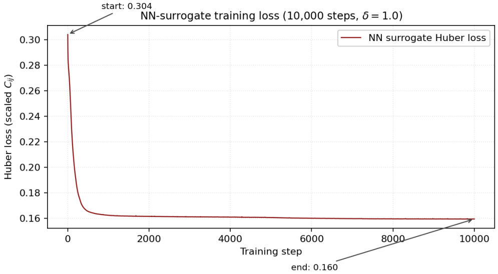
Interactive composition predictor with k-nearest-neighbour distance-based out-of-distribution flagging.

Stage 5b: NN-Driven MEAM Parameter Optimisation

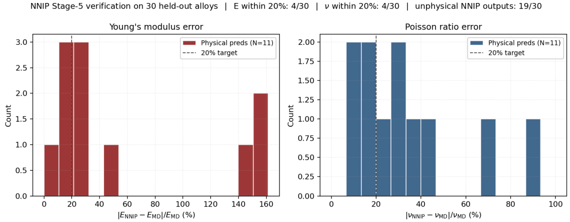
Closed-loop procedure:

1. NN surrogate learns (MEAM params) $\to (C_{11}, C_{12})$.
2. L-BFGS [9] minimises the Huber loss through the surrogate.
3. Optimised parameters are re-evaluated in LAMMPS; new samples are appended and the surrogate retrained.

Atomic checkpointing renders the procedure resumable.



Surrogate Huber-loss decay from 1.0 to 0.16 over 10^4 steps.



Result

Held-out alloys $\to$ pass stability filter	$30 \to 11$
Within 20 % of MD on E / on ν	4 / 4
Median error on E / ν (physical)	23 % / 26 %

30 held-out $\to$ 11 stable $\to$ 4 within tolerance.
19 of 30 violate the Born–Cauchy criterion — the remaining gap motivates the Stage 5b extensions (next slide).

External Validation: Quantitative Summary

By material family ($N = 43$)

Family	N	E MAPE (%) ML	E MAPE (%) MEAM	ν MAPE (%) ML	ν MAPE (%) MEAM
Aluminium	33	2.3	7.4	4.0	4.5
Titanium	6	11.6	140.0	13.2	14.6
Stainless	3	69.6	324.1	75.5	47.3
Ni–Fe (A-286)	1	107.0	—	—	—
Overall	43	**10.8**	42.2	**7.2**	7.1

Aluminium by AA series ($N = 33$)

Al series	N	$\langle E_{\text{lit}} \rangle$ (GPa)	E MAPE (%) ML	E MAPE (%) MEAM
2xxx (Al–Cu)	11	72.7	1.3	8.0
5xxx (Al–Mg)	9	70.5	2.1	7.9
6xxx (Al–Mg–Si)	7	68.9	3.8	4.2
7xxx (Al–Zn–Mg)	6	71.8	2.8	9.3
All aluminium	33	71.1	**2.3**	7.4

Both surrogates reproduce aluminium Young's modulus to within a few percent (ML 2.3 %, MEAM 7.4 %) across all four alloy series; error grows only *off* the Al-rich design domain — steeply for MEAM, gently for the composition NN.

43 basis-covered alloys; MAPE vs experimental data from ASM/MatWeb [10]. Unphysical MEAM runs excluded.

Further Studies

Well-posed optimisation

- Restrict to single-element-anchored cross-terms, ranked by Sobol sensitivity ($\sim$200 params $\rightarrow$ a tractable few)
- Physics-informed loss: Born–Cauchy stability penalties baked into training
- Active-learning expansion to $\geq$500 composition medoids

Beyond the aluminium domain

- Extend the basis to out-of-basis elements (C, V, N) so ferrous and Ti-alloy chemistries are representable
- Broaden MD/DFT sampling into *hcp* (Ti, Mg, Zn) and ferrous (Fe–Cr–Ni) regions
- Symmetry-aware elastic extraction (hexagonal tensor) — removes the cubic-strain error on hcp metals

Reference data & validation

- SQS DFT references [11] at intermediate compositions
- Enlarged DFT set spanning the broadened element domain
- Re-validate each round against ASM/MatWeb experiment — a standing external accuracy loop

The architecture is established; the remaining work is *sampling*, *loss formulation*, and *domain coverage*.

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